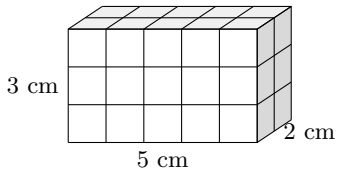
Lesson 1: Volume of Prisms and Cylinders

Assignment

1.



(a) This box is built from 1 cm cubes. How many cubes are in the bottom layer? How many layers? What is the volume?

(b) A box is filled with 1 in cubes. Each layer is 6 cubes by 5 cubes, and there are 4 layers. What is the volume of the box?

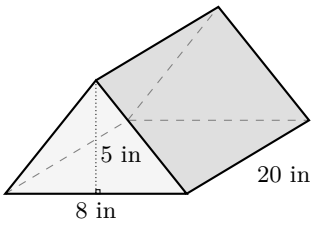
2. Find the volume of each rectangular prism. Round (c) to the nearest tenth.

Object	**Length × width × height**	**Work**	**Volume**
(a) Juice box	6 cm × 4 cm × 10.5 cm		
(b) Microwave box	22 in × 16 in × 14 in		
(c) Moving truck box	4.2 m × 2.1 m × 2.3 m		
(d) Garden storage chest	$4\frac{1}{2}$ ft × 2 ft × $2\frac{1}{4}$ ft		

3. Ashok is choosing a planter for his balcony. One is a cube 40 cm on each edge. The other is a long box 80 cm by 25 cm by 30 cm. Which planter holds more soil, and by how much?

4. Find the volume of each triangular prism.

(a) A wooden doorstop



(b) An A-frame tent. The triangle at the front is 2.4 m wide at the bottom and 1.5 m tall. The tent is 2.8 m long.

5. A triangular prism has a triangle base with $b = 6$ cm and $h = 4$ cm. The prism is 10 cm long. Jordan wrote: $V = 6 \times 4 \times 10 = 240$ cm^3. What mistake did Jordan make? Find the correct volume.

6. Find the volume of each cylinder. Round to the nearest tenth.

(a) radius 3 cm, height 8 cm

(b) diameter 10 in, height 12 in

(c) a hockey puck: diameter 3 in, height 1 in

7. A pillar candle is a cylinder 7.6 cm across and 15 cm tall.

(a) Estimate its volume. Use $\pi \approx 3$ and round the radius to a whole number.

(b) Calculate its volume to the nearest tenth. Is your answer reasonable?

8. Can A is 6.6 cm across and 12.2 cm tall. Can B is 8.4 cm across and 7.6 cm tall. Ella says Can A must hold more because it is taller. Is she right? Show your work.

9. Convert each volume.

(a) 2 ft^3 to in^3

(b) 12 096 in^3 to ft^3

(c) 6 yd^3 to ft^3

(d) 135 ft^3 to yd^3

(e) 3 m^3 to cm^3

(f) 750 000 cm^3 to m^3

(g) Kim says, "1 m = 100 cm, so 1 m^3 = 100 cm^3." Explain her mistake.

10. The space inside a chest freezer is 40 in long, 22 in wide and 30 in deep.

(a) Find the volume in cubic inches.

(b) Change it to cubic feet. Round to the nearest tenth.

(c) The flyer says the freezer holds "15 cu. ft." Is that about right?

11. Destiny is filling a raised garden bed 8 ft long and 4 ft wide with soil 10 in deep. Bags of garden soil hold 1.5 ft^3 and cost $7.97 each.

(a) Find the volume of soil she needs, to the nearest hundredth of a cubic foot.

(b) How many bags should she buy, and what will they cost before tax?

12. A round concrete pad for a fire pit is 12 ft across and 4 in thick. Concrete is sold in half cubic yards at $230.00 per yd^3.

(a) Find the volume of the pad in ft^3, to the nearest hundredth. (Hint: it is a short cylinder.)

(b) How much concrete should be ordered, and what will it cost?

13. A school playground is 12 m by 8 m. It needs wood mulch 20 cm deep. The supplier sells whole cubic yards at $39.00 per yd^3. Use 1 $m^3 \approx 1.31$ yd^3. How many cubic yards should the school order, and what will it cost?

14. Work backwards to find the missing measurement. Check each answer.

(a) A water trough for cattle is a rectangular prism 6 ft long and 2 ft wide. It holds 18 ft^3. How deep is it, in feet and in inches?

(b) An old sandbox is 6 ft by 4 ft, with sand 9 in deep. All the sand is moved to a new sandbox 8 ft by 4.5 ft. How deep is the sand now, in inches?

15. The volume of a box 5 cm long, 4 cm wide and 3 cm high is

A. 12 cm^3 **B.** 60 cm^2 **C.** 94 cm^3 **D.** 60 cm^3

16. A cylinder has a diameter of 10 cm and a height of 5 cm. Its volume is closest to

A. 157 cm^3 **B.** 393 cm^3 **C.** 1571 cm^3 **D.** 79 cm^3

17. Numerical Response A cylinder-shaped water tank is 2 ft across and 3 ft tall. Its volume, to the nearest tenth of a cubic foot, is _____ ft^3.

Lesson 2: Capacity: Litres, Gallons, and US vs British Units

Assignment

1. Convert each measurement.

(a) 3.2 L = _____ mL

(b) 450 mL = _____ L

(c) 1250 cm^3 = _____ L

(d) 0.75 L = _____ cm^3

(e) 4.5 m^3 = _____ L

(f) 600 L = _____ m^3

2. Choose the best unit for each capacity: mL, L or m^3.

(a) one dose of cough syrup _____

(b) a can of house paint _____

(c) the water in a town's water tower _____

(d) a juice box _____

3. A juice box is a rectangular prism 6 cm long, 4 cm wide and 10.5 cm high.

(a) Find its volume in cm^3.

(b) What is its capacity in mL? Can it hold 250 mL of juice?

4. A landscaper carries a box-shaped water tank in her truck to water new trees. Inside, the tank is 1.2 m long, 0.8 m wide and 0.5 m high.

(a) Find the volume of the tank in m^3.

(b) How many litres does the tank hold?

(c) Her pump uses 12 L of water per minute. How long does it take to empty a full tank?

5. A rain barrel is a cylinder with a diameter of 56 cm and a height of 85 cm.

(a) How many litres does the barrel hold? Round to the nearest tenth of a litre.

(b) Joel fills a 9 L watering can from the full barrel. How many full watering cans can he get?

6. Convert each US measurement.

(a) 5 gal = _____ qt

(b) 3 qt = _____ pt

(c) 10 pt = _____ gal

(d) 6 cups = _____ fl oz

(e) 64 fl oz = _____ qt

(f) 2.5 gal = _____ cups

7. At a hockey tournament in North Dakota, the team's water cooler holds 5 US gallons. Each player's bottle holds 20 US fl oz.

(a) How many bottles can be filled from a full cooler?

(b) The team has 18 players, and each player fills a bottle twice in a game. Is one full cooler enough? Explain.

8. (a) How many litres are in 3 US gallons? In 3 British gallons? Round to the nearest hundredth.

(b) How many more litres are in 3 British gallons than in 3 US gallons?

(c) Finish each sentence with "US" or "British".

The _____ pint holds 20 fluid ounces. The _____ gallon is about 3.785 L.

Gas stations in the United States sell gas by the _____ gallon.

9. Hannah's grandfather farms near Swift Current. His old sprayer tank is labelled "200 gallons". It was made in Canada before metric, so these are British gallons.

(a) How many litres does the tank hold?

(b) How many US gallons is that? Round to the nearest tenth.

(c) Explain why the same tank can be called "200 gallons" and "240 gallons".

10. Estimate, then convert. Round to the nearest hundredth (nearest tenth for mL).

(a) 8 US gallons to litres

(b) 45 L to US gallons

(c) a 20 US fl oz bottle of pop to mL

(d) a 1.5 L bottle of water to US fl oz

11. Diego's car uses 7.6 L of gas per 100 km. He plans a 520 km trip in Idaho, where gas costs $3.25 USD per US gallon.

(a) How many litres of gas will the trip use?

(b) How many US gallons is that? Round to the nearest hundredth.

(c) What will the gas cost, in US dollars?

12. Ellie is making muffins from an American recipe. Use 1 US cup $\approx$ 237 mL, 1 tbsp = 15 mL and 1 tsp = 5 mL. Round the cup amounts to the nearest millilitre.

Ingredient	**Work**	**mL**
(a) $2\frac{3}{4}$ US cups of milk		
(b) $\frac{1}{3}$ US cup of oil		
(c) 2 tbsp of maple syrup		
(d) $\frac{1}{2}$ tsp of salt		

(e) If Ellie uses her Canadian 250 mL cup for the milk ($2\frac{3}{4}$ metric cups), how much extra milk is that?

13. A coffee shop makes 60 lattes each morning. Each latte uses 10 US fl oz of milk. Milk comes in 4 L jugs.

(a) How many US fl oz of milk are used each morning?

(b) Change this amount to litres. Round to the nearest tenth.

(c) How many jugs must the shop open each morning?

14. A pet store sells an aquarium labelled "20 gallons (US)". Inside, it measures 61 cm long, 30 cm wide and 41 cm high.

(a) Find its capacity in litres, to the nearest hundredth.

(b) How many litres are in 20 US gallons?

(c) Is the label fair? Explain.

15. Which container holds the most?

A. 1 US gallon **B.** 4 L **C.** 1 British gallon **D.** 9 US pints

16. A bottle of salad dressing holds 750 mL. About how many US fluid ounces is that? (1 US fl oz $\approx$ 29.57 mL)

A. 22 178 fl oz **B.** 25.4 fl oz **C.** 26.4 fl oz **D.** 3.0 fl oz

17. Numerical Response A box-shaped picnic cooler is 45 cm long, 28 cm wide and 30 cm high inside. Its capacity, in litres, is _____ L.

18. Numerical Response A farmer buys 12 US gallons of hydraulic oil. In litres, to the nearest tenth, this is _____ L.

Lesson 3: Temperature: Celsius and Fahrenheit

Assignment

1. Complete the table of referents.

	°C	**°F**
(a) Water freezes		
(b) Water boils		
(c) Normal body temperature		
(d) A comfortable room		

2. Each temperature is missing its unit. Write °C or °F so that it makes sense.

(a) A hot bath: 40 _____

(b) A summer afternoon in Las Vegas: 105 _____

(c) A fresh cup of coffee: 70 _____

(d) A comfortable living room: 70 _____

3. Change each temperature to degrees Fahrenheit.

(a) 30 °C

(b) 5 °C

(c) −10 °C

(d) 15.5 °C

(e) 200 °C

(f) −35 °C

4. Change each temperature to degrees Celsius. Round to the nearest tenth of a degree.

(a) 95 °F

(b) 41 °F

(c) 350 °F

(d) 20 °F

(e) −22 °F

(f) 72 °F

5. Noah changes 77 °F to °C. He types 77 [−] 32 [÷] 1.8 [ENTER] and gets 59.2.
(a) Use the quick estimate to show that his answer cannot be right.

(b) What did he do wrong? Find the correct answer.

6. Bath water for a baby should be between 37 °C and 38 °C. Jesse has a bath thermometer from the US that shows only °F. Between which two °F readings should the water be?

7. On a summer road trip through Utah, a bank sign reads 104 °F.
(a) What is this temperature in °C?

(b) Is it hotter or colder than a 35 °C day in Medicine Hat? By how many Celsius degrees?

8. Marco runs the ice plant at an arena in Red Deer. The manual for the US-made equipment says to keep the ice surface between 22 °F and 26 °F. His display shows the ice is at −4 °C. Is the ice in the right range? Show your work.

9. Use the quick estimate first. Then find the exact answer. Is the estimate good enough?

(a) Tell a friend in Texas that it is 25 °C in Calgary.

(b) Tell a friend in Ohio to set her oven to 220 °C for your pizza recipe.

(c) A website says it is 40 °F in Chicago. Is it cold enough for a winter coat?

10. Which temperature is hotter? By how many Celsius degrees?

(a) 38 °C or 100 °F

(b) −20 °C or −5 °F

11. A chest freezer should be at −18 °C or colder. The freezer at a corner store has a thermometer from the US that reads 5 °F. Is the freezer cold enough? Explain.

12. Overnight in Banff, the temperature falls from 3 °C to −12 °C.

(a) By how many Celsius degrees did it fall?

(b) By how many Fahrenheit degrees did it fall?

(c) Check (b) by changing both temperatures to °F.

13. A US weather report says the temperature rose 36 °F in 3 hours. Kayla says, "$(36 - 32) \div 1.8 \approx 2.2$, so it rose about 2.2 Celsius degrees."

(a) What mistake did Kayla make?

(b) How many Celsius degrees did the temperature rise?

14. On the same winter day it is −12 °C in Calgary and 12 °F in Fargo, North Dakota.

(a) Which city is colder?

(b) By how many Celsius degrees, and by how many Fahrenheit degrees, is it colder?

15. What is 25 °C in degrees Fahrenheit?

A. 77 °F **B.** −3.9 °F **C.** 45 °F **D.** 102.6 °F

16. Which key sequence changes 59 °F to °C correctly?

A. [(] 59 [−] 32 [)] [÷] 1.8 [ENTER]

B. 59 [−] 32 [÷] 1.8 [ENTER]

C. 59 [÷] 1.8 [−] 32 [ENTER]

D. [(] 59 [−] 32 [)] [×] 1.8 [ENTER]

17. During a chinook, the temperature rises by 10 Celsius degrees. By how many Fahrenheit degrees does it rise?

A. 50 **B.** 18 **C.** 42 **D.** 5.6

18. Numerical Response A US deep-fryer manual says to heat the oil to 360 °F. This temperature in degrees Celsius, to the nearest degree, is _____ °C.

Lesson 4: Mass: Grams, Kilograms, Ounces, Pounds and Tons

Assignment

1. Choose a sensible SI unit and a sensible imperial unit for the mass of each object.

Object	**SI unit**	**Imperial unit**
(a) a watermelon		
(b) a hockey puck		
(c) a railway car full of grain		
(d) a dog		

2. Circle the most sensible mass.

(a) a hockey stick:	500 mg	500 g	500 kg
(b) a loaf of bread:	1 oz	1 lb	1 ton
(c) a bull moose:	500 g	500 kg	500 t
(d) a pencil:	6 mg	6 g	6 kg
(e) a big bag of dog food:	30 oz	30 lb	30 tons

3. Convert.

(a) 3.7 kg = _____ g (b) 425 g = _____ kg

(c) 2.05 t = _____ kg (d) 6800 kg = _____ t

(e) 1500 mg = _____ g (f) 0.08 g = _____ mg

(g) 90 g = _____ kg (h) 0.35 kg = _____ g

4. A granola recipe uses 450 g of oats, 180 g of nuts, 120 g of seeds and 90 g of honey. Keisha makes 3 batches.

(a) What is the total mass of the 3 batches, in kilograms?

(b) Oats come in 1 kg bags. How many bags must she buy? How many grams of oats are left over?

5. Old cheddar costs $2.96 per 100 g at the deli counter.

(a) What is the price per kilogram?

(b) How much do 250 g cost?

(c) Holly has $10.00. About how many grams can she buy?

6. Convert.

(a) 7 lb = _____ oz

(b) 2 lb 5 oz = _____ oz

(c) 56 oz = _____ lb

(d) 90 oz = _____ lb _____ oz

(e) 1.75 lb = _____ oz

(f) 4.2 tons = _____ lb

(g) 13 000 lb = _____ tons

(h) 6 oz = _____ lb

7. Find the mistake in each student's work, and give the correct answer.

(a) Tyler wrote: "2.5 lb = 2 lb 5 oz."

(b) Mia wrote: "3 kg = 300 g."

8. A farmer is hauling 18 round bales of hay. Each bale weighs about 1350 lb. His trailer can carry at most 12 tons.

(a) What is the mass of the bales, in tons?

(b) Can he carry all 18 bales in one trip?

(c) What is the greatest number of bales he can carry in one trip?

9. A bakery uses 6 oz of butter in each batch of cookies. Butter comes in 1 lb blocks. The bakery makes 14 batches.

(a) How much butter does it need, in pounds?

(b) How many blocks must it open? How many ounces are left over?

10. Add, subtract or multiply. Write each answer in pounds and ounces.

(a) 4 lb 9 oz + 6 lb 11 oz

(b) 10 lb 2 oz − 3 lb 7 oz

(c) 2 lb 13 oz + 1 lb 6 oz + 15 oz

(d) 3 × (2 lb 9 oz)

11. Baby Leo weighed 8 lb 3 oz when he was born. He lost 9 oz in his first week. By week 3 he had gained 1 lb 4 oz from his lowest weight.

(a) What was his lowest weight?

(b) What did he weigh at week 3?

(c) How much more is that than his birth weight?

12. Shredded mozzarella comes in an 8 oz bag for $5.49 or a 1 lb 4 oz bag for $12.99. Which is the better buy? Show the price per ounce of each.

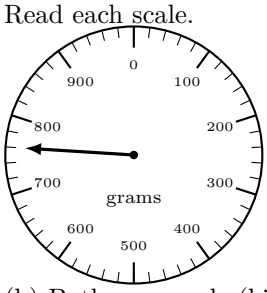
13. Ethan buys a 14 lb 8 oz turkey at $2.40 per lb. After it is cooked, the bones and skin weigh 4 lb 14 oz.

(a) How much did the turkey cost?

(b) How many pounds of meat are there? Write the answer in pounds and ounces and in decimal pounds.

(c) What is the true cost per pound of meat?

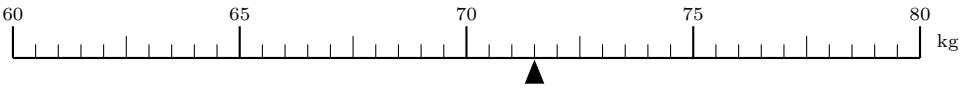
14. Read each scale.



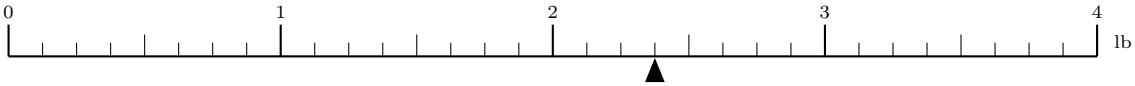
(a) Kitchen scale (grams). One small space: _____

Reading: _____

(b) Bathroom scale (kilograms). One small space: _____ Reading: _____



(c) Produce scale (pounds). Reading: _____



(d) A pot of soil on a scale weighs 2 lb 3 oz. The empty pot weighs 7 oz. How much soil is in the pot?

15. Which mass is the heaviest?

A. 1.8 kg **B.** 1950 g **C.** 0.0017 t **D.** 1 900 000 mg

16. 3 lb 14 oz + 2 lb 5 oz =

A. 6 lb 9 oz **B.** 5 lb 3 oz **C.** 6 lb 3 oz **D.** 7 lb 3 oz

17. 1.25 tons is the same as

A. 1250 lb **B.** 2500 lb **C.** 20 lb **D.** 2250 lb

18. Numerical Response A package of ground beef weighs 45 oz. Its mass in pounds, to the nearest hundredth, is _____ lb.

Lesson 5: Converting Mass Between SI and Imperial

Assignment

1. Circle the heavier mass in each pair. Show how you know.

(a) 1 kg or 2 lb

(b) 3 lb or 1.5 kg

(c) 30 g or 1 oz

(d) 10 tons or 10 t

(e) 500 g or 1 lb

(f) 2 lb 4 oz or 1 kg

2. Estimate in your head. Show your thinking.

(a) a 40 kg dog, in pounds

(b) a 180 lb man, in kilograms

(c) a 10 lb bag of potatoes, in kilograms

(d) 4 oz of cheese, in grams

3. Convert. Round to the nearest tenth.

(a) 15 lb to kg

(b) 45 kg to lb

(c) 8 oz to g

(d) 750 g to oz

(e) 2.5 tons to tonnes

(f) 12 t to tons

4. Find and fix each mistake.

(a) Mateo changed 30 kg to pounds like this: $30 \times 0.454 = 13.6$ lb. Use an estimate to explain why his answer cannot be right. Then find the correct answer.

(b) Zoe wrote: "5 lb 8 oz = 5.8 lb = $5.8 \times 0.454 = 2.6$ kg." Find her mistake and the correct mass in kilograms, to

the nearest tenth.

5. (a) Baby Leo was born at 8 lb 2 oz. Find his mass in kilograms, to the nearest hundredth.

(b) At the clinic, a patient's chart says 72.5 kg. She only knows her weight in pounds. What is it?

(c) A beef roast is 3.6 kg. Find its mass in pounds and ounces, to the nearest ounce.

6. The Nguyen family is flying to Vancouver. Each checked bag may be up to 23 kg. At home their bags weigh 48 lb, 51 lb and 45 lb.

(a) Find the mass of each bag in kilograms, to the nearest tenth. Which bag is over the limit?

(b) Change the 23 kg limit to pounds. Does your answer agree with part (a)?

7. Arjun sells hoodies online. Each hoodie has a mass of 1 lb 5 oz. The post office's prices are below.

Mass of parcel	**Price**
up to 500 g	\$8.25
over 500 g, up to 1 kg	\$11.60
over 1 kg, up to 2 kg	\$15.90
over 2 kg, up to 5 kg	\$22.35

(a) One hoodie goes in a mailer with a mass of 120 g. What is the mass of the parcel in grams? What does it cost to ship?

(b) Two hoodies for the same customer go in one box with a mass of 300 g. What does it cost to ship? How much does Arjun save compared with two separate mailers?

8. Dylan's pickup truck can carry a load of up to 1800 lb, including the driver. Dylan has a mass of 90 kg. Bags of concrete mix are 30 kg each.

(a) Change Dylan's mass and the mass of one bag to pounds.

(b) What is the greatest number of bags Dylan can carry in one trip?

9. Complete the table. Round to the nearest cent.

Item	**Price per lb**	**Work**	**Price per kg**
(a) Chicken breasts	\$4.99		
(b) Bananas			\$1.52
(c) Old cheddar			\$8.80
(d) Gala apples	\$1.79		

10. Salmon fillets cost \$12.99/lb at Store A and \$27.50/kg at Store B.

(a) Which store has the better price? Show your work.

(b) Rami buys 1.5 kg at the better store. How much does he pay?

11. Sliced turkey at the deli costs \$1.89 per 100 g. Hana asks for 2.5 lb for a family lunch. How much will it cost? (Hint: change to grams first.)

12. Liam is making a big batch of macaroni and cheese from an American recipe. It needs 12 oz of shredded cheese and 1.5 lb of macaroni. In his store, cheese comes in 320 g bags and macaroni in 900 g boxes. How many bags and boxes must he buy?

13. At the gym, Ella loads a 20 kg bar with two 45 lb plates and two 10 lb plates.

(a) Find the total mass she is lifting in pounds.

(b) Find the total mass in kilograms, to the nearest tenth.

(c) Ella's goal is to lift 80 kg. About how many more pounds of plates does she need?

14. A mass of 5 kg is about

A. 2.27 lb **B.** 10 lb **C.** 11 lb **D.** 2.5 lb

15. Pork chops are on sale for \$4.40/lb. The price per kilogram is

A. \$2.00 **B.** \$8.80 **C.** \$6.60 **D.** \$9.68

16. Which mass is the greatest?

A. 2 kg **B.** 4.5 lb **C.** 70 oz **D.** 1950 g

17. Numerical Response A bag of flour has a mass of 3 lb 12 oz. Its mass in kilograms, to the nearest hundredth, is _____ kg.

Lesson 6: From Volume to Mass: Bushels, Tonnes and Conversion Factors

Assignment

1. Find each mass.

Material	Factor	Volume	Work	Mass
(a) Water	1 kg per L	3.5 L		
(b) Water	1 g per mL	600 mL		
(c) Maple syrup	1.33 kg per L	4 L		
(d) Diesel in a farm tank	0.84 kg per L	1100 L		
(e) Concrete	2.4 t per m^3	3.5 m^3		

(f) A bucket of water has a mass of 9.5 kg. About how many litres of water are in it?

(g) A hot tub holds 2 m^3 of water. What is the mass of the water in tonnes?

2. Topsoil has a mass of about 1.3 t per m^3. Mei orders 5 m^3 of topsoil for a garden. Her trailer can carry at most 2 t.

(a) What is the mass of the topsoil?

(b) How many cubic metres can the trailer carry in one load? Round down to the nearest tenth.

(c) How many loads will Mei need?

3. Here are the factors for four more crops.

Crop	Bushels per tonne
Field peas	36.7
Flax	39.4
Canola	44.1
Oats	64.8

(a) Find the mass of one bushel of flax, to the nearest tenth of a kilogram.

(b) List the crops from heaviest to lightest per bushel.

_____

(c) Finish the sentence. A bushel of oats is lighter than a bushel of peas because _____

_____

4. Change tonnes to bushels. Round to the nearest bushel.

(a) 8 t of wheat (36.7 bu/t)

(b) 20 t of canola (44.1 bu/t)

(c) 3.2 t of oats (64.8 bu/t)

(d) 15.5 t of barley (45.9 bu/t)

5. Change bushels to tonnes. Round to two decimals.

(a) 500 bu of wheat (36.7 bu/t)

(b) 1000 bu of barley (45.9 bu/t)

(c) 2400 bu of oats (64.8 bu/t)

(d) 75 bu of canola (44.1 bu/t)

6. Kayla wants to know the mass of 640 bushels of canola (44.1 bu/t). She writes: $640 \times 44.1 = 28\,224$ t.

(a) How can you tell, without a calculator, that her answer is wrong?

(b) What did she do wrong? Find the correct mass.

7. A horse stable near Okotoks buys 90 bushels of oats (64.8 bu/t).

(a) Find the mass in tonnes (two decimals) and in kilograms (nearest kg).

(b) Find the mass in pounds. Use 1 kg $\approx$ 2.2 lb.

(c) The stable's trailer can carry 3500 lb. Can it take all the oats in one trip?

8. A semi-trailer grain truck (a "Super B") may carry at most 42 t of grain.

(a) Can it carry 1600 bushels of wheat (36.7 bu/t)? Show your work.

(b) What is the most canola (44.1 bu/t) it can carry, in whole bushels?

9. Change each price. Round to the nearest cent.

(a) wheat (36.7 bu/t): \$290.00/t to \$/bu

(b) canola (44.1 bu/t): \$15.80/bu to \$/t

(c) oats (64.8 bu/t): \$300.00/t to \$/bu

(d) barley (45.9 bu/t): \$5.40/bu to \$/t

10. Jesse is selling 2200 bushels of canola (44.1 bu/t). The elevator offers \$700.00 per tonne. A crushing plant offers \$15.90 per bushel.

(a) Change the plant's price to dollars per tonne. Which offer is better?

(b) How much will Jesse get from each buyer? How much more is the better offer?

11. A farm near Vulcan grew wheat (36.7 bu/t) on 160 acres. The field yielded 55 bushels per acre.

(a) How many bushels did the farm harvest?

(b) What is the mass of the wheat, to the nearest tonne?

(c) What is the wheat worth at \$7.90 per bushel?

12. A round bin has a diameter of 21 ft. It holds barley (45.9 bu/t), level and 18 ft deep. Use 1 bu $\approx$ 1.28 ft^3.

(a) Find the volume of the barley, to the nearest tenth of a cubic foot.

(b) How many bushels is that?

(c) What is the mass, to the nearest tenth of a tonne?

13. A metric grain bin has a diameter of 5 m. It holds oats (64.8 bu/t), level and 4.5 m deep. Use 1 m^3 $\approx$ 27.5 bu.

(a) Find the volume of the oats, to two decimals.

(b) How many bushels is that, to the nearest bushel?

(c) What is the mass, to the nearest tenth of a tonne?

14. Aaron must haul 4500 bushels of barley (45.9 bu/t) to the elevator. His truck can carry 14 t per trip.

(a) Find the mass of the barley, to two decimals.

(b) How many trips does he need? Explain why the answer is not 7.

15. Challenge. A standard bushel of canola weighs 50 lb. One tonne is about 2204.6 lb. Show where the factor 44.1 bu/t comes from.

16. A farmer delivers 920 bushels of canola (44.1 bu/t). The mass of the canola is about

A. 20.86 t **B.** 40 572 t **C.** 0.05 t **D.** 2.09 t

17. Which crop is the **heaviest** per bushel?

A. corn, 39.4 bu/t **B.** soybeans, 36.7 bu/t **C.** sunflower seed, 73.5 bu/t **D.** mustard seed, 44.1 bu/t

18. Numerical Response An elevator pays $298.00 per tonne for oats (64.8 bu/t). The price per bushel, to the nearest cent, is $_____.

Lesson 7: Unit 4 Review

Assignment

This review is set up like the unit test. Show your work on the written questions. Use the conversion numbers on the summary page unless a question gives you others.

Part A: Written Response

1. Fill in the blanks.

(a) 4.5 L = _____ mL (b) 2500 g = _____ kg (c) 3.2 m^3 = _____ L

(d) 650 cm^3 = _____ mL (e) 0.8 t = _____ kg (f) 1250 mg = _____ g

2. Fill in the blanks.

(a) 3 gal = _____ qt (b) 6 cups = _____ pt (c) 2 lb 5 oz = _____ oz

(d) 5000 lb = _____ tons _____ lb (e) 3 ft^3 = _____ in^3 (f) 54 ft^3 = _____ yd^3

3. Convert each temperature. Round to the nearest tenth of a degree where needed.

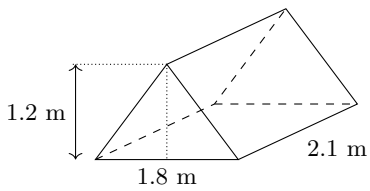
(a) 30 °C to °F (b) 14 °F to °C (c) −40 °C to °F (d) 425 °F to °C

4. A shipping box is 45 cm long, 30 cm wide and 25 cm high.

(a) Find its volume in cubic centimetres.

(b) What is its capacity in litres?

5. A camping tent is shaped like a triangular prism. The front is a triangle 1.8 m wide and 1.2 m high, and the tent is 2.1 m long.



(a) Find the area of the triangle at the front.

(b) Find the volume of air inside the tent.

6. A can of paint is a cylinder 16.5 cm across and 19 cm tall.

(a) Find the volume of the can, to the nearest cubic centimetre.

(b) Change the volume to litres.

(c) The label says the can holds 3.78 L of paint (one US gallon). Explain why the can's volume is a bit more than that.

7. Ella is filling a sandbox 6 ft long and 5 ft wide with sand 10 in deep.

(a) Find the volume of sand in cubic feet.

(b) Is that more or less than 1 cubic yard?

(c) Bags of play sand hold 0.5 ft^3. How many bags does she need?

8. A pudding recipe from a British cookbook uses 2 pints of milk. Ayesha's measuring jug is in millilitres.

(a) How many millilitres is 2 British pints?

(b) How many millilitres would 2 US pints be? (Hint: 1 US gal = 8 pt.)

(c) About how much more milk is in the British amount? Is that about the 20% you expected?

9. Marco's pickup has a 98 L gas tank. He is driving to Montana, where gas costs \$3.49 USD per US gallon. In Calgary, gas costs \$1.45 per litre. Use 1 USD = 1.38 CAD.

(a) How many US gallons does the tank hold?

(b) Find the Montana price per litre in US dollars, then in Canadian dollars.

(c) Where is gas cheaper? About how much does Marco save filling an empty tank there?

10. A butcher cuts three roasts that weigh 4 lb 11 oz, 3 lb 14 oz and 5 lb 7 oz.

(a) What is the total weight in pounds and ounces?

(b) She sells 9 lb 13 oz of the meat. How much is left?

11. Convert. Round to the nearest tenth unless told otherwise.

(a) A 25 kg bag of dog food, in pounds.

(b) 8 oz of cheese, in grams.

(c) A newborn of 3.4 kg, in pounds and ounces (to the nearest ounce).

(d) A farm truck rated for 2.5 tons, in tonnes (to the nearest hundredth).

12. Chicken thighs cost \$4.49/lb at Store A and \$9.29/kg at Store B.

(a) Change Store A's price to a price per kilogram. (Use 1 kg $\approx$ 2.2 lb.)

(b) Which store is cheaper? How much do 3 kg cost there?

13. A farmer near Red Deer harvests 6885 bushels of barley. For barley, use 45.9 bu/t.

(a) How many tonnes is that?

(b) A buyer offers \$4.75 per bushel. What is the whole crop worth? What is that price per tonne?

(c) A grain truck carries 38 t. How many truck loads does the crop need?

14. A round grain bin is 21 ft across. It is filled with barley to a depth of 14 ft. Use 1 bushel $\approx 1.28\ \text{ft}^3$, and for barley 45.9 bu/t.

(a) Find the volume of the barley in cubic feet.

(b) About how many bushels is that?

(c) About how many tonnes?

Part B: Multiple Choice

15. Which is the most reasonable mass for a loonie?

A. 7 g **B.** 7 kg **C.** 70 mg **D.** 0.7 t

16. It is 20 °C outside. The quick estimate gives about

A. 70 °F **B.** 52 °F **C.** 40 °F **D.** −7 °F

17. A can is 10 cm across and 12 cm tall. Its volume is about
A. 942.5 cm^3 **B.** 3769.9 cm^3 **C.** 377.0 cm^3 **D.** 300 cm^3

18. A hot tub holds 1.5 m^3 of water. How many litres is that?
A. 15 L **B.** 150 L **C.** 1500 L **D.** 1 500 000 L

19. A landscaper orders 3 yd^3 of topsoil. How many cubic feet is that?
A. 9 ft^3 **B.** 27 ft^3 **C.** 81 ft^3 **D.** 36 ft^3

20. Which statement about gallons is true?

A. A US gallon and a British gallon are the same size.

B. A British gallon is about 20% bigger than a US gallon.

C. A US gallon is about 20% bigger than a British gallon.

D. A British gallon is about 4.546 mL.

21. A bag of rice weighs 2.5 lb. How many ounces is that?
A. 40 oz **B.** 25 oz **C.** 32 oz **D.** 30 oz

22. Grapes cost \$2.99/lb. Using 1 kg $\approx$ 2.2 lb, their price per kilogram is about
A. \$1.36 **B.** \$6.58 **C.** \$2.99 **D.** \$5.98

Part C: Numerical Response

23. Numerical Response A juice box is a rectangular prism 6.5 cm by 4 cm by 10.5 cm. Its capacity, in millilitres, is _____ mL.

24. Numerical Response Normal body temperature is 37 °C. Changed to degrees Fahrenheit and rounded to the nearest tenth, it is _____ °F.

25. Numerical Response A hockey bag full of gear weighs 34 lb. Using 1 lb $\approx$ 0.454 kg, its mass in kilograms, to the nearest tenth, is _____ kg.

26. Numerical Response A farmer sells 2400 bushels of wheat. For wheat, use 36.7 bu/t. The number of tonnes, to the nearest tenth, is _____ t.
